

ME 300: Engineering Thermodynamics

Summer 2023

Instructor: Dr. Jacqueline O'Connor
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The best way to get me outside office hours is by e-mail – jxo22@psu.edu. Please do not use Canvas email – I will respond much more promptly if you use my PSU email account.

Course book: Turns, S. R., & Pauley, L. L. (2020). *Thermodynamics: concepts and applications*. Cambridge University Press.

Course modality: This is a fully online course and except for the three exams will not have required class meetings.

Office Hours: TBD by vote

Pre-requisites/Co-requisites: CHEM 110 and MATH 141

COURSE DESCRIPTION: This course covers thermodynamics concepts, properties of pure substances, and the first and second law analysis of systems and control volumes.

The objective of this course is to prepare engineering students to effectively solve theoretical and applied thermodynamics problems. Multiple forms of assessment will be used throughout the course in order to evaluate student learning. In addition, multiple professional skills will be practiced and demonstrated throughout the course, such as engineering communication and teamwork, since these skills are important in all engineering professions.

By the end of this course, students will be able to:

1. Obtain thermodynamic data necessary to solve thermodynamic problems and when necessary use appropriate approximations. These skills include the use of equations of state and/or tabulated property tables.
2. Write the First Law of Thermodynamics in both appropriate forms for both closed system and control volume problem.
3. Solve problems requiring First Law analysis that produce a simple single answer.
4. Make appropriate assumptions when applying the First Law to a “real-world” problem.
5. Write the Second Law of Thermodynamics in the appropriate forms for both closed system and control volume problem.
6. Apply the Second Law to determine the performance limitations of a given thermodynamic system.
7. Apply thermodynamic concepts to describe the performance of the individual components of an engineering system, e.g. a power plant, a jet engine, and then relate that information to the overall performance of the entire system.
8. Physically interpret and apply integrals and derivatives to solve thermodynamic problems.
9. Translate complex word problems into an orderly and logical problem-solving approach.

Dr. O'Connor reserves the right to alter or change this syllabus at any point in the semester. If this should occur, Dr. O'Connor will post an updated syllabus on Canvas and email it to the entire class. It is each student's responsibility to check their email and Canvas frequently, as well as to pay attention in class to stay up to date on any changes.

COURSE PHILOSOPHY: This is a 12-week course on engineering thermodynamics in an asynchronous delivery method. The course is loosely designed based on the theory of competency-based learning, which is centered around achieving understanding for each student. Assignments are designed in both their structure and execution to act primarily as learning tools, not assessments. In particular, homework, quizzes, and "explained examples," will be used to provide student with ample opportunity to practice and master concepts before being assessed on them in three exams throughout the semester. The course also incorporates many aspects of [Universal Design for Learning](#), which provide students with many avenues by which to learn and practice the material. Lectures are presented in short videos with activities built into the video, and the same material is also provided in annotated slides and the textbook. Some of these aspects of the course may feel new, but they have been designed using evidence-based practices and student learning at the core. Dr. O'Connor would like to acknowledge Dr. Andrea Gregg, Assistant Teaching Professor in Mechanical Engineering and an expert in online pedagogy, for her help with the design of this course. Additionally, the images in the videos, if not otherwise stated, are subject to Creative Commons Usage.

While this is an online course, I hope to create a community to support our learning. Some assignments will be interactive, and Dr. O'Connor will be available for office hours each week. Study groups are strongly encouraged, and while most of the assessments will be personal, a collegial environment is most representative of the working situations in which modern engineers will find themselves. Use the introduction videos in the first discussion board to help meet your classmates!

COMMUNICATION POLICIES: Because part of becoming a professional in engineering is learning to effectively communicate, I expect that all elements of communication between students and with me as an instructor be conducted with respect. All emails sent to me must subscribe to professional expectations for communication, including a salutation, clear emails in full sentences that are grammatically correct and have been spell-checked, communicate clearly how you would like me to assist you, and end with a respectful closing followed by your name. I request that you call me Dr. O'Connor or Professor O'Connor, and address me as such in emails and in person. If your emails with me do not subscribe to these requirements, I will kindly request that you revise your emails and re-send it to me before I will respond to your email.

You are welcome to email me at any time, however, I will be most responsive to emails during business hours. Although you may receive emails from me on weekends/holidays/evenings, I reserve the right to not be "on call" outside of working hours. All emails should include ME 300 in the subject line, as well as the subject of the email in order so I can quickly scan my emails. I will respond within one full business day to emails. Please do NOT use the email through Canvas – I cannot check that as often.

TYPES OF ASSIGNMENTS

- **Homework:** Homework is due weekly on Fridays (except on exam weeks). The best method to approaching the homework is to do as much as you can on your own, then attend office hours, post on the discussion board, or speak with your peers. Students are responsible for the material in the homework and so working the material out on your own will maximize your learning. Exam questions will closely resemble homework questions.
- **Explained Examples:** Three times during the semester, each student will be paired with another student and will complete an “explained example.” These assignments are based on a fantastic study by Brent and Felder (original paper is available on Canvas) that showed that students developed deeper understanding of concepts by explaining worked examples to each other, rather than just watching the professor do the example.

In the first part of this assignment, students will record a Zoom meeting with each other and explain the example step by step, switching presenters as indicated in the assignment. Students will turn in the recording, graded for completion.

The second part of this assignment requires students to review the video of Dr. O’Connor solving and explaining the same example. Students will then write a reflection comparing their step-by-step explanation to that of Dr. O’Connor’s. This will be submitted the following week. Given the dynamic nature of the engineering profession, learning *how to learn* is an essential skill for success. This part of the assignment emphasizes metacognitive reflection and gives students an opportunity to reflect on their own thinking processes, identifying both strengths and areas in need of improvement.

- **Quizzes:** Quizzes will be online in Canvas and are *individual efforts*. These are short open-book, open-note check-ins on the material covered in the videos and the book to ensure that the student understands the basics. Students are able to take each quiz three times, and the final score will be the average of the attempts. The questions may change from one attempt to the next, but the same concepts will be quizzed in each question each time.
- **Exams:** Exams are *individual efforts* and will be held three times in the semester, dates below; they will be administered synchronously on Zoom. These are open-book, open-note exams that will last 1.5 hours, with 15 minutes after the exam finishes to upload written work. Exams will include both concept questions and worked problems; practice exams will be provided with solution ahead of the exam. The final exam is not cumulative, although each concept in the course builds on previous concepts, and so in a sense, all exams are cumulative. I take academic integrity during exams very seriously, which is why the exam is proctored and all students will have to sign an academic integrity pledge at the beginning of the exam. However, I do not think that the use of AI-based proctoring is fair or conducive to a productive test-taking environment. As such, I trust that you all will take the integrity of the exam seriously with Zoom.

If you are unable to attend these exams for University-approved reasons, please notify Dr. O’Connor within the first week of the course.

- **Exam 1:** Thursday, June 22, 2023, 6:00 – 7:30 PM, Eastern time zone
- **Exam 2:** Thursday, July 20, 2023, 6:00 – 7:30 PM, Eastern time zone
- **Exam 3:** Friday, August 11, 2023 at assigned final exam time
- **Zoom link for exams:**
<https://psu.zoom.us/j/94085813985?pwd=bmY0cm1lTnBwdTJmSUhGVnBZNc9rUT09>
Password: 487733

GRADING:

Homework (drop lowest):	25%
Explained Examples:	15%
Quizzes (drop two lowest):	20%
Exams:	40%

Late drops prior to first exam	-WN
Late drops after first exam:	
With a score $\geq 60\%$	-WP
With a score $< 60\%$	-WF

GRADING SCALE

A	93
A-	90
B+	87
B	83
B-	80
C+	77
C	70
D	60

TIPS FOR VIRTUAL LEARNING: The university has a number of resources for virtual learning:

- Advice for engineering students, *from* engineering students: <https://www.engr.psu.edu/assets/docs/virtual-learning-tips.pdf>
- IT issues (half-way down the page, on the right): <https://keeplearning.psu.edu/>
- Remote learning FAQ: <https://keeplearning.psu.edu/remote-learning-support/>

CONTESTING GRADES: We are all human, and there may be grading mistakes from time to time. If you have an issue with how an assignment was graded, please provide a formal application for grade change, including a copy of the particular question and your original answer, and a paragraph explaining why you believe you deserve a grade change. These applications should be emailed to me NO EARLIER than 24 HOURS after the assignment was returned; verbal requests for grade change will not be considered. If there was an obvious mistake in grading, I will immediately correct the issue. If the grade change is more subjective in nature, I will file your request and reconsider at the end of the semester if a change in this grade could change your final grade in the class.

LATE ASSIGNMENTS: Assignments that are submitted late are subject to a 15% grade deduction in the first 24 hours, and 50% after the first 24 hours. If students know an assignment will be late for a University approved reason, please contact Dr. O'Connor ahead of the due date to determine a new due date. Note that one homework and two quizzes are dropped in the semester.

MISSED QUIZZES/EXAMS: No make-up quizzes or exams will be given except as required by University policy. See me **prior** to any anticipated absence, preferably at the beginning of the semester.

COUNSELING AND PSYCHOLOGICAL SERVICES: Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The university offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. These services are provided by staff who welcome all students and embrace a philosophy respectful of clients' cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

- [Counseling and Psychological Services at University Park \(CAPS\)](http://studentaffairs.psu.edu/counseling/) (http://studentaffairs.psu.edu/counseling/): 814-863-0395
- Counseling and Psychological Services at [Commonwealth Campuses](http://senate.psu.edu/faculty/counseling-services-at-commonwealth-campuses/) (http://senate.psu.edu/faculty/counseling-services-at-commonwealth-campuses/)
- Penn State Crisis Line (24 hours/7 days/week): 877-229-6400
Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741

EDUCATIONAL EQUITY AND REPORTING BIAS: Consistent with University Policy AD29, students who believe they have experienced or observed a hate crime, an act of intolerance, discrimination, or harassment that occurs at Penn State are urged to report these incidents as outlined on the University's Report Bias webpage (<http://equity.psu.edu/reportbias/>)

STATEMENT ON ACCOMODATIONS FOR STUDENTS WITH DISABILITIES: Penn State welcomes students with disabilities into the University's educational programs. Every Penn State campus has an office for students with disabilities. The Student Disability Resources (SDR) website provides contact information for every Penn State campus: <http://equity.psu.edu/sdr/campus-contacts>. For further information, please visit Student Disability Resources: <http://equity.psu.edu/sdr>.

In order to receive consideration for reasonable accommodations, you must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: <http://equity.psu.edu/sdr/guidelines>. If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with an accommodation letter. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. You must follow this process for every semester that you request accommodations.

STATEMENT ON ACADEMIC INTEGRITY

Academic dishonesty will not be tolerated *at all*. I hope that everyone can develop enough pride in his or her own work and abilities that this will never be a problem. When you earn an Engineering degree from Penn State, the University is certifying that you are capable of performing engineering duties at a professional level. Course grades are the sole basis on which the College of Engineering certifies your degree with the assumption that your course grades are a valid assessment of your own knowledge and abilities. If you have cheated, you have falsified that credential. Therefore, we must have academic integrity expectations to ensure the validity of your grade and your degree. Detailed information on this topic can be found at <https://www.engr.psu.edu/faculty-staff/academic-integrity.aspx>. Some examples are given below:

COPYING ON TEST: Copying in a complicit manner with another student; passing answers via text or email; discussing answers in exam; having someone else complete your exam, etc.

PLAGIARISM: The fabrication of information and citations; submitting others work from professional journals, books, articles and papers; submission of other students papers or lab results or project reports and representing the work as one's own; fabricating in part or total, submissions and citing them falsely, etc.

ACTS OF AIDING OR ABEADING: Facilitating acts by others; unauthorized collaboration of work; permitting another to copy from exam; writing a paper for another; inappropriately collaborating on home assignment or exam without permission or when prohibited, etc.

UNAUTHORIZED POSSESSION: Of examinations, through purchase or supply; stealing exams; failing to return exams on file; selling exams; photocopying exams; buying exams; any possession of an exam without the custodian's permission, etc.

SUBMITTING PREVIOUS WORK: Submitting a paper, case study, lab report or any assignment that had been submitted for credit in a prior class without the knowledge and permission of the instructor.

TAMPERING WITH WORK: Changing own or another students work product such as lab results, papers, or test answers; tampering with work either as a prank or in order to sabotage another work, etc.

GHOSTING: Taking a quiz, an exam, performing a laboratory exercise or similar evaluation in place of another; having another take a quiz, an exam, or perform an exercise or similar evaluation in place of the student, etc.

ALTERING EXAMS: When instructor returns graded exams for in class review and subsequently collects them, student changes incorrect answers and seeks favorable grade adjustment asserting that instructor made mistake in grading; other forms may include changing the letter or and/numerical grade on test; obtaining test in discretely, etc.

Course Schedule

Week 1: May 15 – 19, 2023		Videos: M1.1 – M1.5, M2.1 Book chapters: 1.1 – 1.6, 3.1
Friday, May 19	Homework 1: covers pre-knowledge, study modules, and videos M1.1 – M1.5	
Sunday, May 21	Office Hours Poll Introduction Discussion Board	
Week 2: May 22 – 26, 2023		Videos: M2.2 – M2.8 Book chapters: 1.5, 2.1 – 2.3
Monday, May 22	Study Smarter Training: 12:00 – 2:00 PM online Quiz 1: covers videos M1.1 – M1.5	
Thursday, May 25	Quiz 2: covers videos M2.1 – M2.4	
Friday, May 26	Homework 2: covers videos M2.1 – M2.6	
Week 3: May 29 – June 2, 2023		Videos: M3.1 – M3.5 Book chapters: 2.4 – 2.5
Monday, May 29	Quiz 3: covers videos M2.5 – M2.9	
Thursday, June 1	Quiz 4: covers videos M3.1 – M3.3 Group 1: EE 1 Video	
Friday, June 2	Homework 3: covers videos M2.7 – M3.3	
Week 4: June 5 – 9, 2023		Videos: M3.6 – M3.8, M4.1 – M4.6 Book chapters: 2.6 – 2.9
Monday, June 5	Quiz 5: covers videos M3.4 – M3.5	
Tuesday, June 6	Group 1: EE 1 Reflection	
Thursday, June 8	Quiz 6: covers videos M3.6 – M3.8 Group 2: EE 2 Video	
Friday, June 9	Homework 4: covers videos M3.4 – M4.4	
Week 5: June 12 – 16, 2023		Videos: M5.1 – M5.7 Book chapters: 4.1 – 4.3, 5.1
Monday, June 12	Quiz 7: covers videos M4.1 – M4.6	
Tuesday, June 13	Group 2: EE 2 Reflection	
Thursday, June 15	Quiz 8: covers videos M5.1 – M5.3 Group 3: EE 3 Video	
Friday, June 16	Homework 5: covers videos M4.5 – M5.4	
Week 6: June 19 – 23, 2023		Videos: M5.8, M6.1 – M6.5 Book chapters: 3.2 – 3.4, 5.2 – 5.3, 8.2 – 8.3
Monday, June 19	Quiz 9: covers videos M5.4 – M5.8	
Tuesday, June 20	Group 3: EE 3 Reflection	
Thursday, June 22	Group 1: EE 4 Video Exam 1, 6:00 – 7:30 PM, ET on Zoom: covers videos M1.1 – M5.8	
Week 7: June 26 – June 30, 2023		Videos: M6.6 – M6.10 Book chapters: 8.4 – 8.6
Monday, June 26	Quiz 10: covers videos M6.1 – M6.3	
Tuesday, June 27	Group 1: EE 4 Reflection	
Thursday, June 29	Quiz 11: covers videos M6.4 – M6.7	

	Group 2: EE 5 Video
Friday, June 30	Homework 6: covers videos M5.5 – M6.5
Week 8: July 3 – 7, 2023	Videos: M7.1 – M7.6 Book chapters: 6.1 – 6.4
Monday, July 3	Group 2: EE 5 Reflection
Tuesday, July 4	Independence Day
Thursday, July 6	Quiz 12: covers videos M6.8 – M6.10 Group 3: EE 6 Video
Friday, July 7	Homework 7: covers videos M6.6 – 7.4
Week 9: July 10 – 14, 2023	Videos: M7.7 – M7.13 Book chapters: 6.2d, 7.1 – 7.5
Monday, July 10	Quiz 13: covers videos M7.1 – M7.5
Tuesday, July 11	Group 3: EE 6 Reflection
Thursday, July 13	Quiz 14: covers videos M7.6 – M7.10 Group 1: EE 7 Video
Friday, July 14	Homework 8: covers videos M7.5 – M7.11
Week 10: July 17 – 21, 2023	Videos: M7.14 – M7.15, M8.1 – M8.4 Book chapters: 6.2, 7.5
Monday, July 17	Quiz 15: covers videos M7.11 – M7.15
Tuesday, July 18	Group 1: EE 7 Reflection
Thursday, July 20	Group 2: EE 8 Video Exam 2, 6:00 – 7:30 PM ET on Zoom: covers videos M6.1 – M7.15
Week 11: July 24 – 28, 2023	Videos: M8.5 – M8.8 Book chapters: 9.1, 9.6
Monday, July 24	Quiz 16: covers videos M8.1 – M8.4
Tuesday, July 25	Group 2: EE 8 Reflection
Thursday, July 27	Quiz 17: covers videos M8.5 – M8.8 Group 3: EE 9 Video
Friday, July 28	Homework 9: covers videos M7.12 – M8.4
Week 12: July 31 – August 4, 2023	Videos: M8.9 – M8.15 Book chapters: 9.2 – 9.5
Monday, July 31	Quiz 18: covers videos M8.9 – M8.12
Tuesday, August 1	Group 3: EE 9 Reflection
Thursday, August 3	Quiz 19: covers videos M8.13 – M8.15 Bonus EE 10 Video Bonus EE 11 Video
Friday, August 4	Homework 10: covers videos M8.5 – M8.15
Week 13: August 7 – 11, 2023	
Monday, August 7	Bonus EE 10 Reflection Bonus EE 11 Reflection
Friday, August 11	Final Exam: covers videos M8.1 – M8.15